3D WEBSOFT INC. FOUNDATION

SCHOOL FUTURE TECH PROGRAM

Faculty Name :

Course Duration : 5 Months (150 Days)

Classes : Offline & Online

Daily Teaching : 3 Hours

Total Hours : 300 Hours

LEVEL 3 –Future Engineers(9th–10th Standard)

MODULE 1 – Python Foundations (Classes 1–10)

Class 3 – Installing Python & Setting Up VS CODE

Complete Theory & Practical Notes

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Learning Objectives

By the end of this class, the session aims to:

1. Introduce students to the Python programming environment.
2. Explain the purpose of **Python**, **VS Code**, and the **Python extension**.
3. Familiarize students with the basic steps involved in setting up a Python development environment.
4. Guide students through installing Python on their computer.
5. Teach students how to verify that Python has been installed correctly.
6. Introduce students to the VS Code interface and its main features.
7. Guide students in installing and configuring the Python extension in VS Code.
8. Teach students how to select the appropriate Python interpreter.
9. Enable students to create and save their first Python (.py) file.
10. Introduce students to running Python programs and viewing output in the terminal.
11. Develop basic troubleshooting skills for common installation and configuration issues.
12. Build students' confidence in using a professional programming environment.

Session

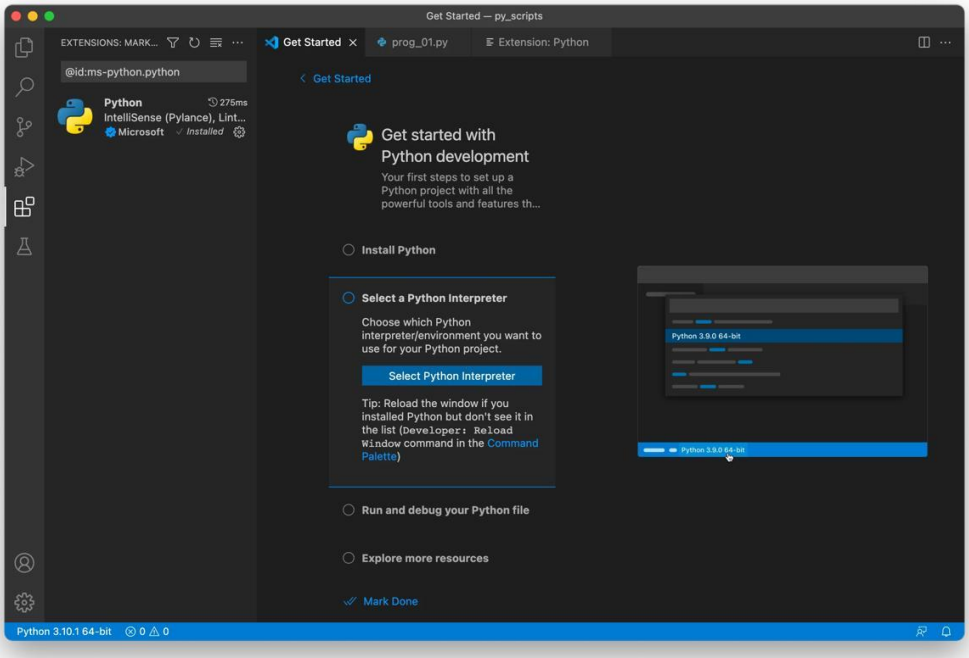
Topic	Type
Welcome & What Are We Setting Up?	Theory
Python vs VS Code	Theory
Installing Python	Demonstration
Verify Python Installation	Practical
Installing VS Code	Demonstration + Practical
Break / Setup Challenge	Activity
Install Python Extension	Practical
Select Interpreter & Create Workspace	Practical
Create & Run First Python Program	Practical
Troubleshooting Challenge	Activity
Quiz + Exit Check	Assessment

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1. WELCOME — WHAT ARE WE BUILDING TODAY?







Today we are going to prepare your computer to become your Python laboratory.

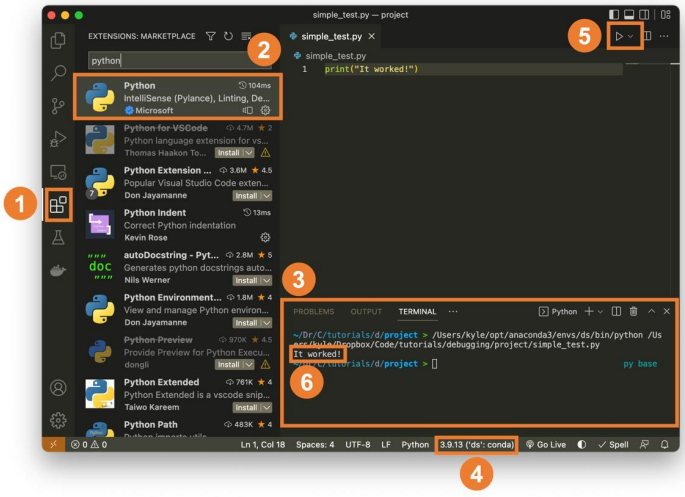
- Where do programmers write code?
- What software do we need to write Python?
- Can VS Code run Python by itself?
- What happens when we click "Run"?

Today's class is about setting up the **development environment**.

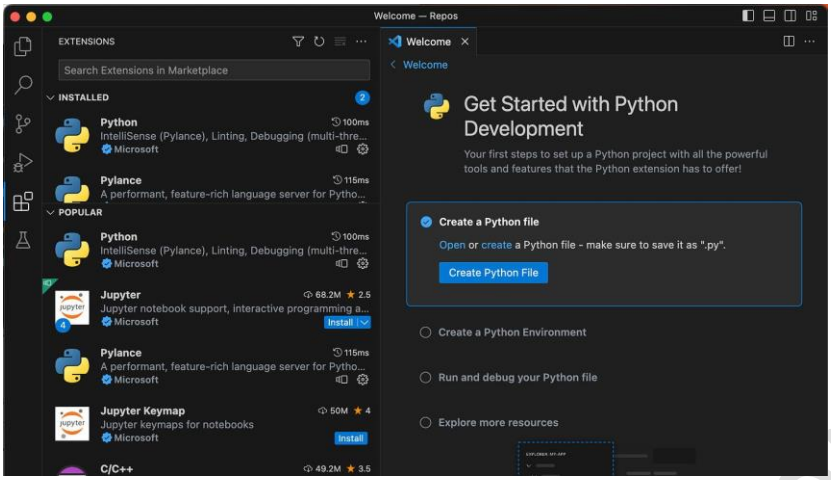
2. PYTHON vs VS CODE

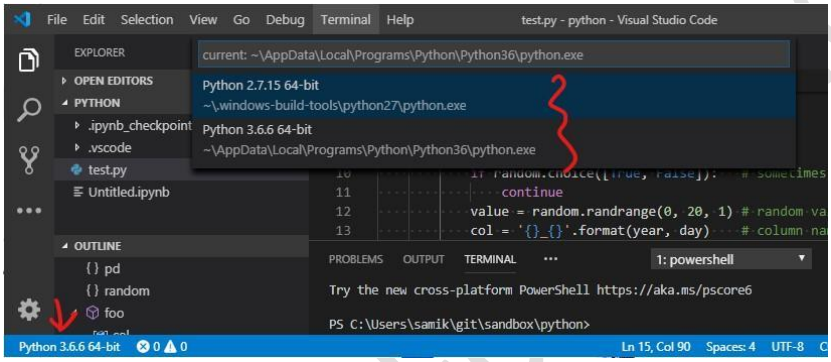
THEORY

This distinction is very important for beginners.

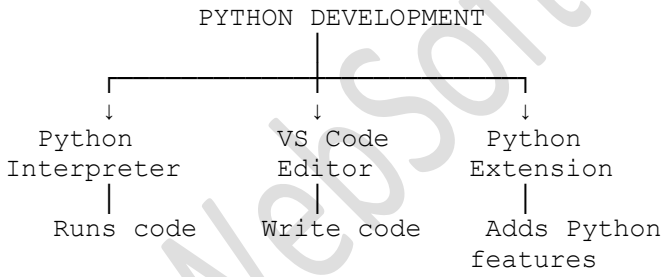


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Write on the board:



Python Interpreter

The program that actually executes Python code.

VS Code

The application where we write and organize our code.

Python Extension

Adds Python-specific features such as code assistance, running, debugging and interpreter selection to VS Code.

VS Code's official documentation describes these as three separate components.

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Simple Analogy

Tell students:

> ***Think of a kitchen.***

VS Code = Kitchen

Where you prepare your food.

Python = Chef

The person/program that actually performs the cooking.

Python Extension = Kitchen Equipment

Tools that make the cooking process easier.

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3. INSTALLING PYTHON

THEORY

Explain:

Before VS Code can run Python programs, the computer needs a Python interpreter.

For a standard beginner setup, use the official Python downloads. The current Python release line is available from Python.org; as of this class preparation, Python 3.14.7 is listed as the latest Python 3 release.

Official resource

Python Downloads — Official Python.org

PRACTICAL

Windows students

1. Open the official Python download page.
2. Download the current supported Python installer.
3. Start the installation.
4. Follow the installer instructions.
5. Complete the installation.

The exact installer interface can vary between Python releases, so demonstrate using the current installer available on the day of class.

For macOS, use the current official macOS installer or an appropriate package-management approach. Microsoft's current VS Code Python tutorial also notes that macOS system Python should not be treated as the supported development installation.

4. VERIFY PYTHON INSTALLATION

This is an essential practical exercise.

Open the terminal / command prompt.

Ask students to try:

```
python --version
```

If their system uses the python3 command, try:

```
python3 --version
```

The purpose is to confirm that a Python interpreter is available.

Expected result

Something similar to:

```
Python 3.x.x
```

The exact version will depend on the version installed.

PRACTICAL ACTIVITY

"Show me your Python version!"

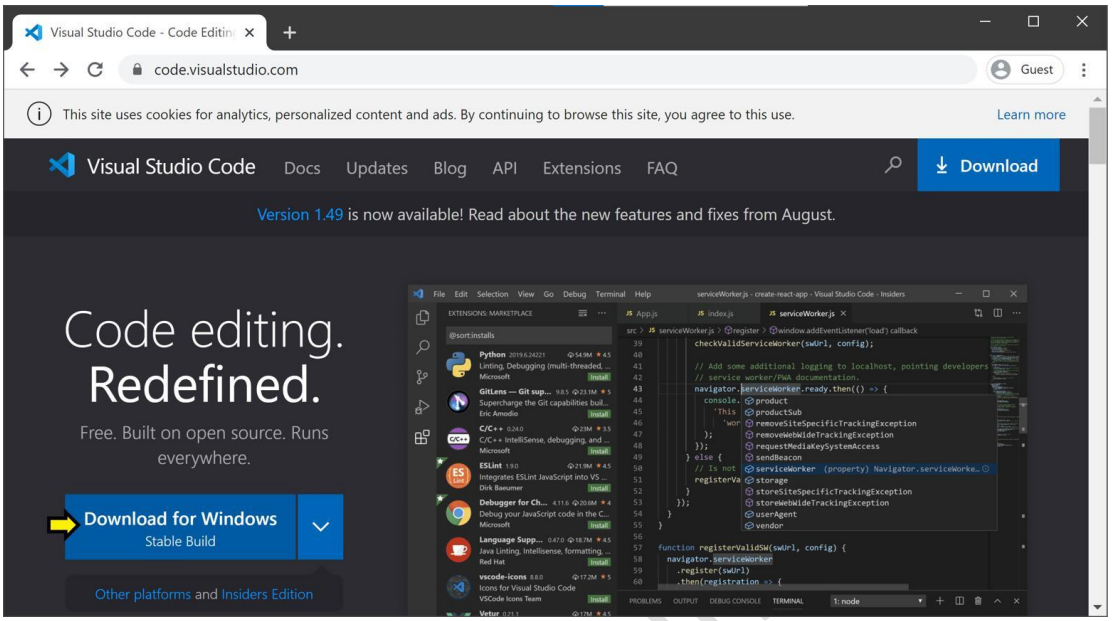
Each student must show the teacher:

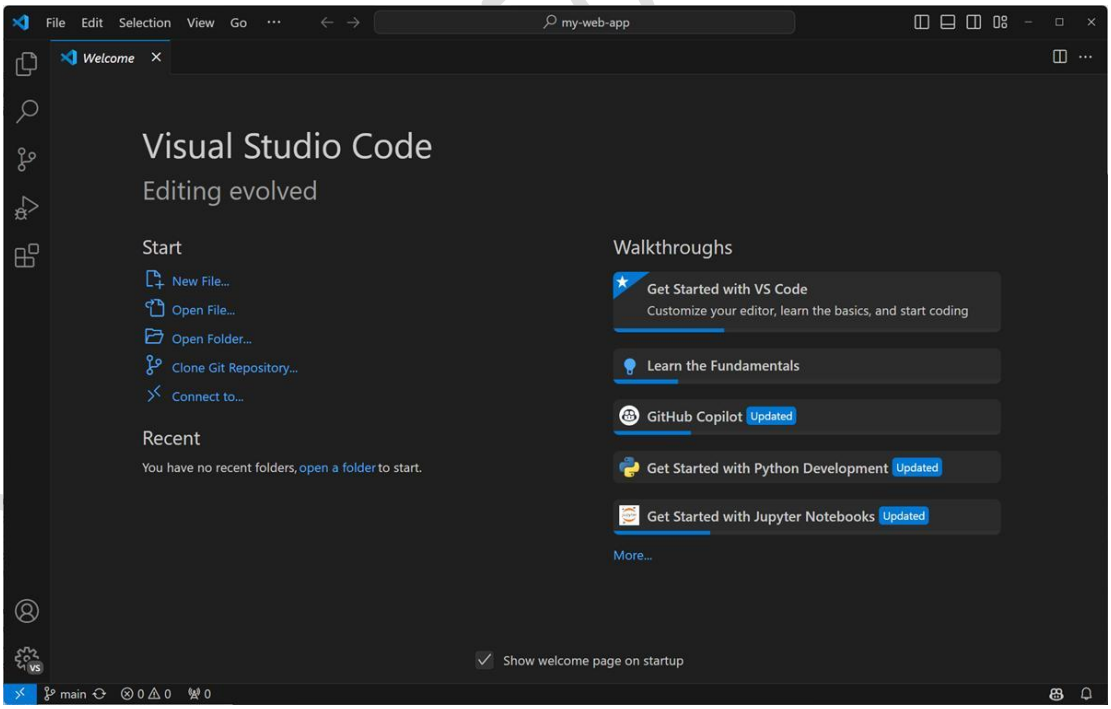
```
Python 3.x.x
```

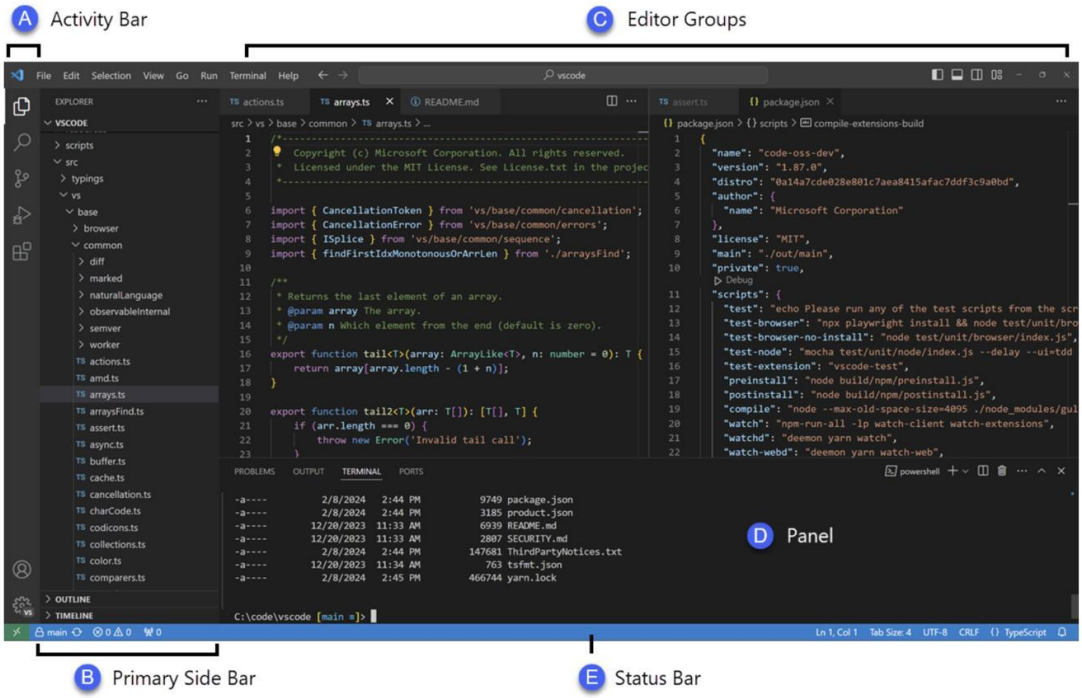
- ✓ Python installed
- ✓ Python version displayed
- ✓ No installation error

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6. INSTALLING VISUAL STUDIO CODE







VS Code is the environment in which we will write, organize, run and debug our Python programs.

Official resource

Visual Studio Code — Official Website

Practical Steps

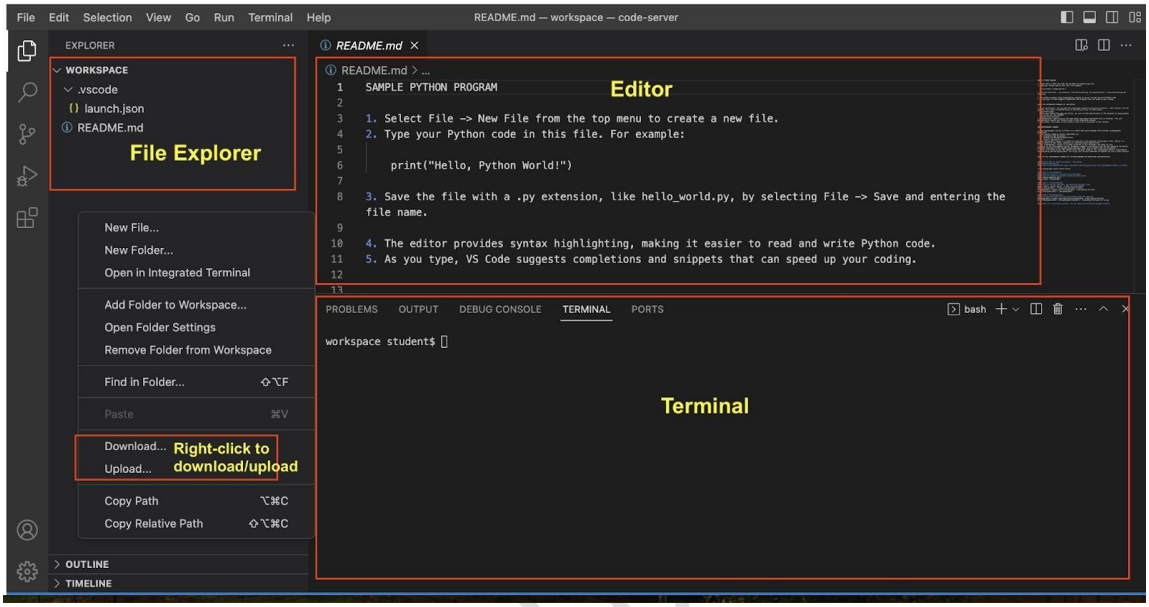
1. Open the official VS Code website.
2. Download the installer for the student's operating system.
3. Install VS Code.
4. Open VS Code.
5. Look at the main interface.

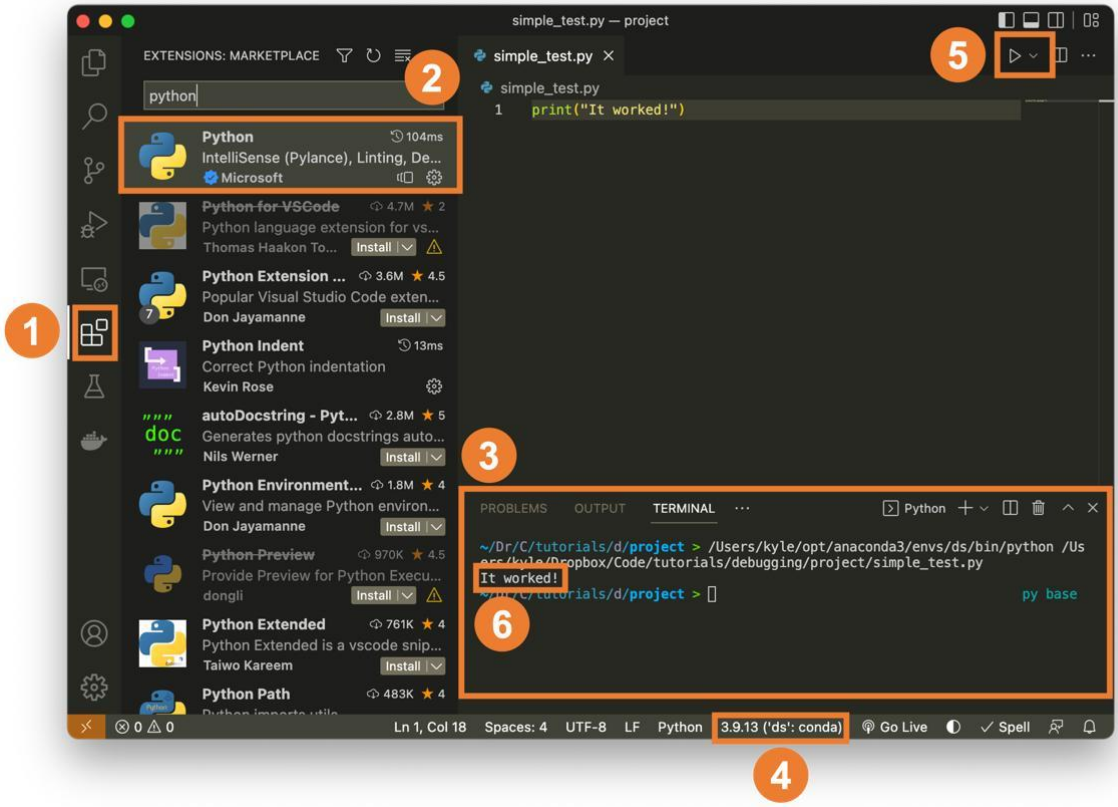
The official VS Code Python documentation recommends installing VS Code separately from Python and then adding Python support through the Python extension.

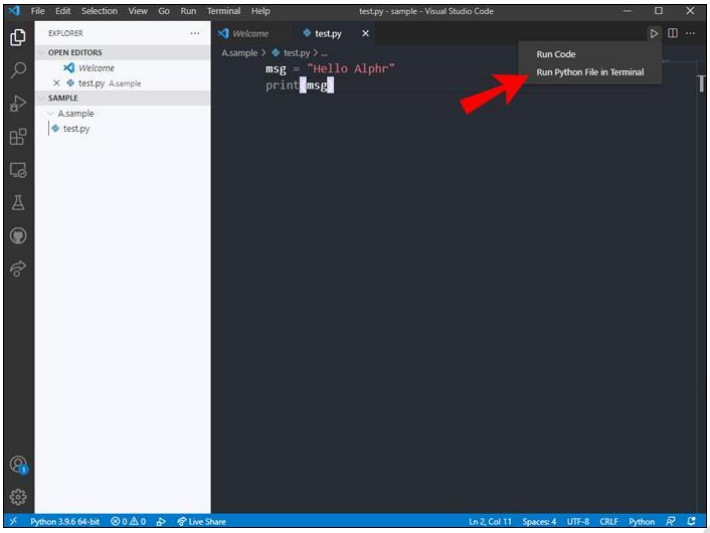
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6. VS CODE INTERFACE TOUR

Once everyone has VS Code open, introduce:







Important areas

Area	Purpose
Explorer	Files and folders
Editor	Write code
Extensions	Install extensions
Terminal	Run commands
Run button	Execute programs
Status bar	Shows useful information

Today we only need the tools necessary to create and run Python.

SETUP CHALLENGE

Give students a short break.

After the break, announce:

SETUP RACE

Students must complete:

- ✓ `Python installed`
- ✓ `Python verified`

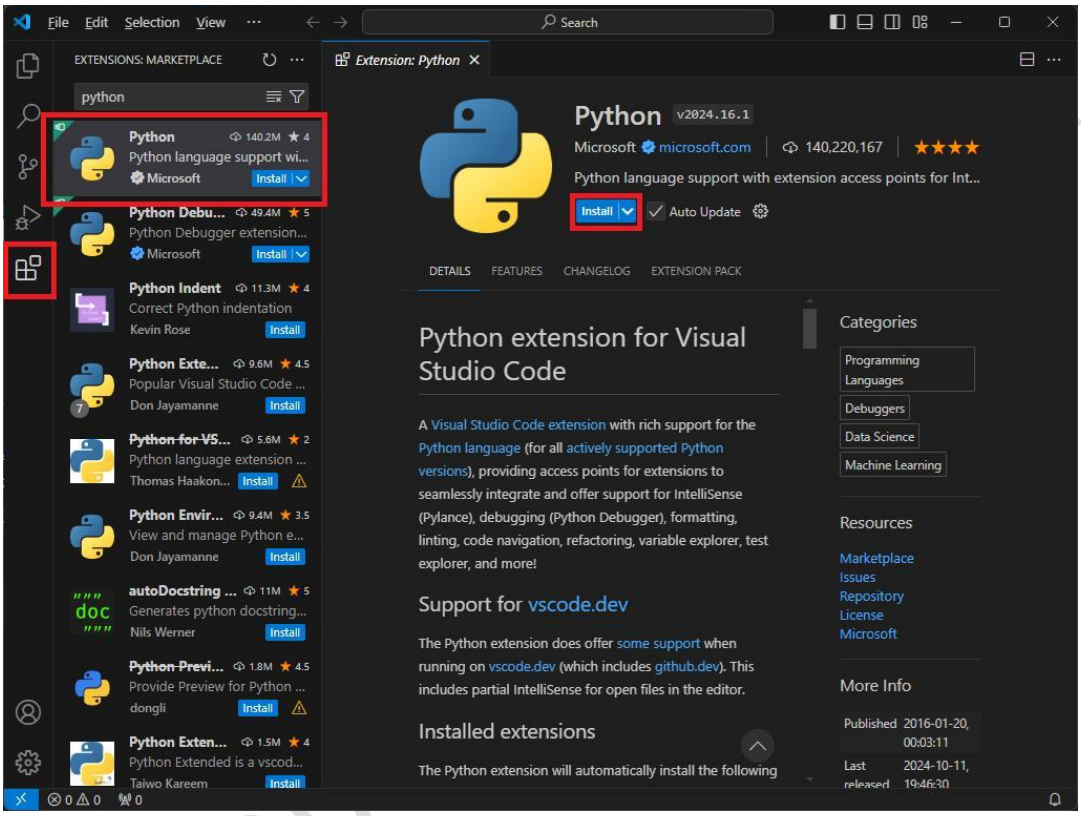
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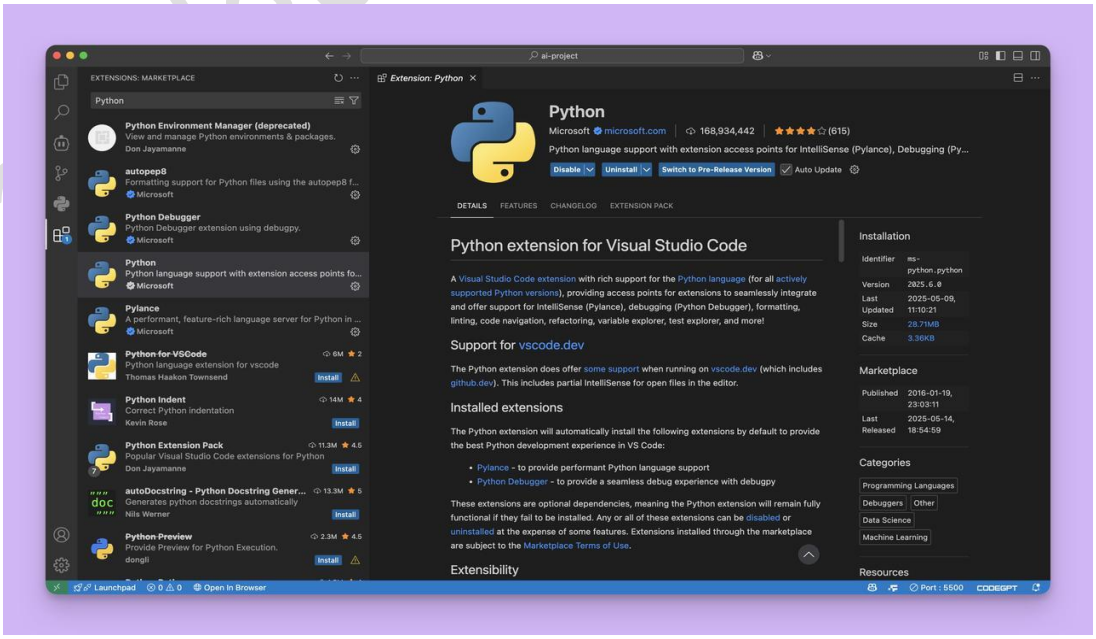
- ✓ VS Code installed
- ✓ VS Code opened

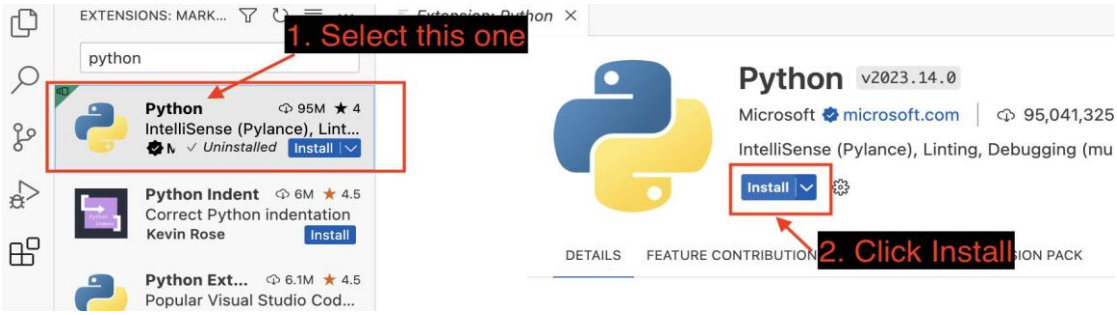
Students who finish help their partner **without touching the partner's computer**.

This encourages peer learning.

7. INSTALL THE PYTHON EXTENSION







VS Code is a general-purpose editor. The Python extension gives VS Code Python-specific capabilities.

Practical Steps

1. Open VS Code.
2. Open Extensions**.**
3. Search for Python.
4. Select the Python extension from Microsoft**.**
5. Install it.

The official VS Code documentation recommends installing the Python extension from the VS Code Marketplace.

8. CREATE A PYTHON PROJECT FOLDER

Professional programmers don't keep all their files randomly on the desktop. We organize our projects into folders.

Create:

```
Python_Foundations
```

Inside it, you eventually have:

```
Python_Foundations
│
├── Class01
│
├── Class02
│
└── Projects
```

For today's class:

```
Python_Foundations
└── Class01
```

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9. SELECT THE PYTHON INTERPRETER

This is a critical concept.

VS Code needs to know which Python installation it should use.

In VS Code, use the **Python: Select Interpreter** command and select the Python interpreter that you installed. The official VS Code documentation specifically recommends selecting the interpreter after installing Python.

Board diagram

```
    VS CODE
       ↓
 Python Extension
       ↓
Select Interpreter
       ↓
Python Installation
       ↓
  Run Program
```

10. CREATE YOUR FIRST PYTHON FILE

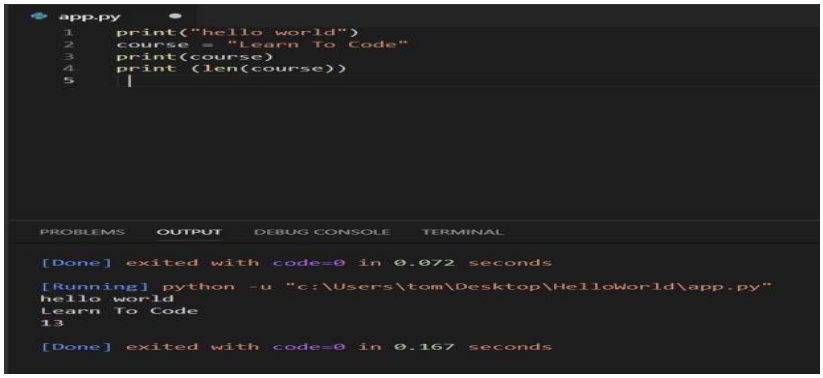
Create a new file:

`hello.py`

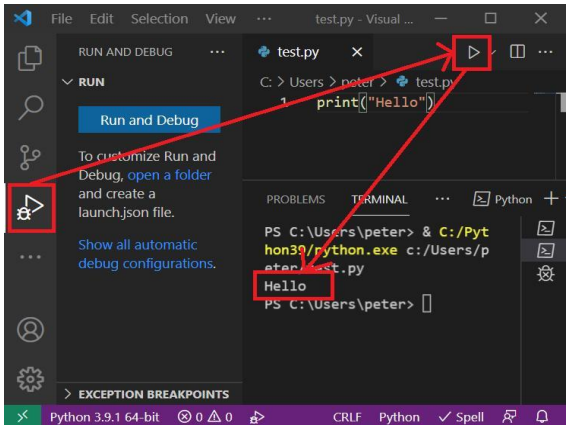
.py is the standard file extension for a Python source file.

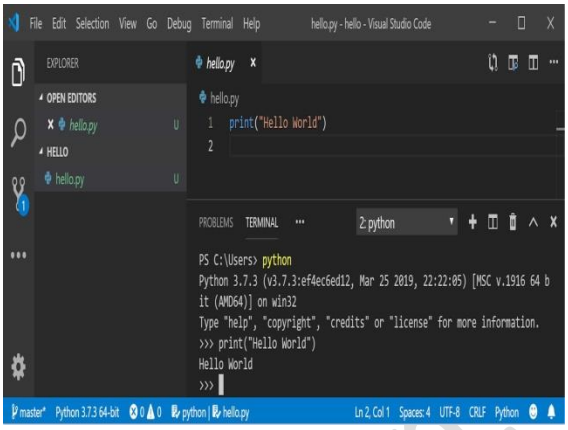
```
print("Hello, Future Engineer!")
```

11. RUN YOUR FIRST PYTHON PROGRAM



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Students run the program using VS Code's Python run functionality.

Expected output:

```
Hello, Future Engineer!
```

Congratulations! Your computer is now ready for Python programming.

12. PRACTICAL CHALLENGE

Students modify the program:

```
print("Hello, Future Engineer!")
print("My name is _____")
print("I am learning Python")
print("I love technology")
print("My dream is to become _____")
```

Run it.

Expected output

```
Hello, Future Engineer!
My name is Ananya
I am learning Python
I love technology
My dream is to become a Robotics Engineer
```

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13. TROUBLESHOOTING ACTIVITY

Introduce three common beginner problems

Problem 1 — Python not found

Possible message:

```
'python' is not recognized...
```

- Is Python installed?
- Is the correct interpreter selected?
- Does the system use `python3`?

Problem 2 — VS Code doesn't recognize Python

Check:

```
Python Extension
      ↓
   Installed?
      ↓
Python Interpreter
      ↓
   Selected?
```

Problem 3 — Program doesn't run

Check:

1. Is file saved?
2. Is file named .py?
3. Is Python interpreter selected?
4. Is the Python extension installed?
5. Is there a syntax error?

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14. FINAL ASSESSMENT

Quick Quiz

Q1. What is Python?

Answer: A programming language and interpreter/runtime ecosystem used to execute Python programs.

Q2. What is VS Code?

Answer: A code editor/development environment used to write and manage code.

Q3. What does the Python extension do?

Answer: It adds Python-specific development features to VS Code.

Q4. What does `.py` mean?

Answer: A Python source-code file.

Q5. What does this code do?

```
print("Hello")
```

Answer: Displays Hello.

Q6. What is a Python interpreter?

Answer: The program that executes Python code.

Learning Outcomes

By the end of the class, students will be able to:

1. Explain what Python is and why it is used in programming.
2. Describe the role of VS Code as a code editor and development environment.
3. Differentiate between the Python interpreter, VS Code, and the Python extension.
4. Install Python successfully on their computer.
5. Verify the Python installation using the terminal/command prompt.
6. Install and open Visual Studio Code.
7. Install the Python extension in VS Code.
8. Select the correct Python interpreter in VS Code.
9. Create and save a Python file with a .py extension.
10. Write a simple Python program using the print() function.
11. Run a Python program from VS Code.
12. View and interpret the program output in the terminal.
13. Identify and resolve basic Python setup and execution problems.
14. Demonstrate readiness to use Python and VS Code for the upcoming programming classes.

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